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CIVIL ESTIMATION & STRUCTURAL BOQ REPORT

IS 456:2000 & IS 13920:2016 Compliant Engineering Takeoff • Date: 31 Oct 2026

**REF: BQ-105829**

Jurisdiction: Bengaluru, Karnataka

## Green Villa Estate

Client: Maaz Ayoob | Location: Bengaluru, Karnataka | Package: STANDARD PACKAGE | Structure: RCC Framed

| PLOT & SITE GEOMETRY | CONSTRUCTED BUILT-UP (SBA) | PACKAGE RATE BENCHMARK |
|---|---|---|
| Dimensions: 55 ft × 55 ft | Total SBA (BUA): 1,800 sq.ft | Tier: Standard Package |
| Gross Plot Area: 3,025 sq.ft | Carpet Area (70%): 2,117.5 sq.ft | Agreed Rate: Rs. 2,250 / sq.ft |
| Area in Sq.Yards: 336.1 sq.yd | Floor Count: 2 Levels (G + 1) | GST Status: Statutory 18% Applicable |
| Orientation: North Facing | Typology: Independent Residential | Methodology: Deterministic IS 456 |

## 1. FINANCIAL & STATUTORY TAX SUMMARY

| Financial Component | | |
|---|---|---|
| Constructed Built-up Area (SBA / BUA) | Gross chargeable area across all levels | **1,800 sq.ft** |
| Selected Construction Package Tier | Comprehensive material & workmanship scope | **Standard Package** |
| Agreed Package Internal Base Rate | Pre-GST rate per sq.ft constructed area | **Rs. 2,250 / sq.ft** |
| Base Construction Cost (SBA × Package Rate) | 1,800 sq.ft × Rs. 2,250 / sq.ft | **Rs. 40,50,000** |
| Statutory GST (18%) | Statutory tax on works contract (IS / GST Law) | **Rs. 7,29,000** |
| **GRAND TOTAL ESTIMATED INVESTMENT** | **Total all-inclusive contract investment** | **Rs. 47,79,000** |
| Effective All-Inclusive Cost per Sq.Ft | Comprehensive post-tax cost per sq.ft SBA | **Rs. 2,655 / sq.ft** |

## 2. FLOOR-WISE AREA BREAKDOWN

| # | | | | | |
|---|---|---|---|---|---|
| 1 | Ground Floor | 630 sq.ft | 900 sq.ft | 4 | 50.0% |
| 2 | First Floor | 630 sq.ft | 900 sq.ft | 4 | 50.0% |
| | **Total Constructed Coverage** | **2,117.5 sq.ft** | **1,800 sq.ft** | **8** | **100.0%** |

## 3. STRUCTURAL MATERIAL TAKEOFF (IS 456:2000 SPECIFICATIONS)

| Material Category | | | |
|---|---|---|---|
| **Reinforcement Steel Rebar** | JSW Neosteel Fe 550D High-Ductility TMT (IS 1786) | **6.30** | Tonnes |
| **Structural Grade Cement** | UltraTech Super 53 Grade / PPC WeatherPro (IS 12269) | **756** | Bags (50kg) |
| **Manufactured Sand (M-Sand)** | Manufactured Double-Washed VSI M-Sand & P-Sand (IS 383) | **3,420** | Cu.Ft |
| **Coarse Aggregate Jelly** | Machine-Crushed Blue Metal Granite (20mm & 12mm) | **2,430** | Cu.Ft |
| **Wall Masonry Units** | Autoclaved Aerated Concrete (AAC) Lightweight Blocks | **2,250** | Blocks / Sq.Ft |
| **Flooring & Wall Tiling** | Double Charged 4x2 GVT Vitrified Tile Slabs | **2,266** | Sq.Ft |
| **Waterproofing System** | Dr. Fixit 2K Polymer Elastomeric Membrane (Toilets & Terrace) | **1.0** | Package (L.S) |

## 4. STATUTORY & ENGINEERING COMPLIANCE NOTES

1. Structural Engineering Standards: Quantities calculated deterministically complying with IS 456:2000 (RCC), IS 13920:2016 (Ductile Detailing), and IS 1786:2008 (High Strength Deformed Steel Bars).

2. Geotechnical Soil Assumption: Safe Bearing Capacity (SBC) of foundation soil is benchmarked at 180-200 kN/m² for isolated pad footings. Deep piles or heavy rocky strata are evaluated via site borehole tests.

3. Comprehensive Workmanship Scope: Includes mechanized batch mixing, cube test verification (M25 grade 28-day compressive strength), immersion vibration, and statutory 14-day water curing protocols.

4. GST & Statutory Taxes: Statutory 18% Goods and Services Tax (GST) is itemized. External municipal sanction charges, betterment fees, and utility deposits (BESCOM / BWSSB) are client responsibilities.

*Authoritative Legal Notice: This document is an algorithmic civil engineering and quantity surveying estimation issued by Buildiqo.ai. Final contractor agreement may vary based on architectural elevation detailing, approved structural drawings, and site-specific topography.*